Ore Geochemistry from Hyperspectral Images: an Honest Baseline on HIDSAG

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Abstract

The feed of a semi-autogenous grinding (SAG) mill is characterised by sampling batches of ore and analysing them in the laboratory; hyperspectral imaging of such samples could estimate their composition faster and at lower cost than chemical assays. We test this idea on the public HIDSAG GEOCHEM data set (Ehrenfeld et al., 2023a): 28 samples of porphyry copper ore, imaged in the visible to short-wave infrared (400 nm to 2500 nm) and assayed for 18 elements by portable X-ray fluorescence. Because every image of a sample carries the same label, the only honest test is on samples the model has never seen. Under this test, a principal-component ridge regression on the mean spectrum of each image reduces the error of the trivial mean predictor by 29 % on average over the nine major elements, and by about half for Ca and Fe; a 1D convolutional network does worse than this linear model. The camera sees minerals, not elements: Al and K follow the Al–OH absorption of muscovite, Ca the molecular-water band, and Cu is predicted through its association with alteration minerals. Grain size changes the absorption depths more than composition does. The near-perfect accuracy published for this data set cannot be reproduced with any split of the data that we tried. Code and data: Chavarría Morales (2026).

1 Motivation

The throughput and the energy consumption of a SAG mill depend on properties of its feed such as hardness, moisture and clay content. These properties are not observed continuously: batches of ore are sampled, and each sample is characterised by a set of estimation techniques, from chemical assays to metallurgical tests. Geometallurgy links the results to the mineralogy and the chemistry of the ore. Among these techniques, reflectance spectroscopy is fast, non-destructive and cheap per sample; if it can estimate the chemistry of a sampled batch, it can complement or partly replace slower assays, and its estimates can serve as inputs to models of the mill. This report is the first step of that chain: *how well can hyperspectral images predict the chemical composition of an ore sample?* The second step, from composition and spectra to the variables of the mill, is outside its scope; it is discussed briefly in Section 8.

The project was done for a course on machine vision with a free choice of topic. Its first version, a Jupyter notebook, trained a convolutional network on 28 mean spectra and evaluated it on a random split. This report is a re-analysis that asks three questions the notebook did not ask:

1. What is an honest estimate of the prediction error, and is it better than chance?
2. What does the camera physically see, and what in the images is not ore?
3. Why are the accuracies published for this data set so much better?

2 Materials

HIDSAG (Ehrenfeld et al., 2023a,b) contains five subsets of geometallurgical samples from Chilean copper deposits, each imaged with two push-broom cameras: a VNIR camera (942 bands, 400 nm to 1000 nm) and a SWIR camera (268 bands, 1000 nm to 2500 nm), co-registered at the resolution of the SWIR camera. We use the subset GEOCHEM: 28 crushed samples of SAG feed, each split into a fine and a coarse size fraction and photographed in two crops per fraction, 106 crops of about 70×150 pixels in total. Each sample has one pXRF analysis of 18 elements. We study the nine elements whose variation across samples is clearly above the detection limit (Al, Si, P, S, K, Ca, Ti, Fe, Cu); Zn, As, Pb, Mo and the other traces are reported in the repository.

The unit of independence is the sample. The 106 crops and their 1.1×10^6 pixels share 28 labels. The amount of information about the relation between spectrum and composition is therefore set by the 28 samples, not by the number of pixels. A model that is tested on a pixel, a patch or a crop of a sample that is also in its training set can succeed by recognising the sample instead of predicting its composition (Roberts et al., 2017; Kaufman et al., 2012).

Spectral features. The raw counts are divided by 10^4. The VNIR spectrum is averaged into 10 nm bins between 420 and 1000 nm (58 features) to remove its noise; the SWIR spectrum is kept between 1000 and 2450 nm (259 features). Each crop is summarised by its mean spectrum (*crop level*) or by 40 means of 25 random pixels (*patch level*), drawn from 1000 pixels stored per crop by the extraction script. Figure 1 shows the spectra of the samples with the lowest and the highest Ca content.

3 What the camera sees

Visible and short-wave infrared light does not interact with elements but with chemical bonds and electronic transitions in minerals (Hunt, 1977): vibrations of OH and H_2O near 1400 and 1900 nm, Al-OH near 2200 nm (muscovite, kaolinite), Fe/Mg-OH near 2300 nm (chlorite, biotite), and charge transfer of ferric iron below 550 nm. Elements can therefore be predicted only to the extent that they reside in, or co-vary with, minerals that have such absorptions.

Figure 2 shows the correlation of the continuum-removed reflectance with each element across the 28 samples, and Table 1 the same for five absorption band depths. The clearest pattern is that of muscovite (sericite alteration): Al and K go with a deeper Al–OH band at 2205 nm and a deeper OH band at 1405 nm, and K with a shallower Fe/Mg–OH band ($r = -0.67$). Ca goes with a deeper molecular-water band at 1935 nm ($r = 0.49$), consistent with gypsum or hydrated clays, but only weakly with the sulfate band of gypsum at 1745 nm. Cu and Fe have no diagnostic absorption in this range; Cu-rich samples absorb more between 1700 and 1880 nm and at the OH band, i.e. copper is predicted through the alteration that accompanies the mineralisation, not through copper minerals. With 28 samples, $|r| > 0.37$ is needed for significance at the 5 % level, and a few hundred bands are tested: single correlations are fragile, and the best single band generalises poorly (Section 5).

Grain size dominates the absorption depths. The last two columns of Table 1 carry the most important result of this section. For the water band at 1935 nm and the Al–OH band at 2205 nm, the coarse fraction of a sample absorbs more than its fine fraction by 3.5 times the standard deviation between samples, consistently in all 25 pairs. This is the expected physics (longer optical paths in larger grains), but it means that a feature extracted from an image depends more on the particle size distribution than on the composition. Unless the sample preparation fixes the size fraction, this effect has to be modelled or measured.

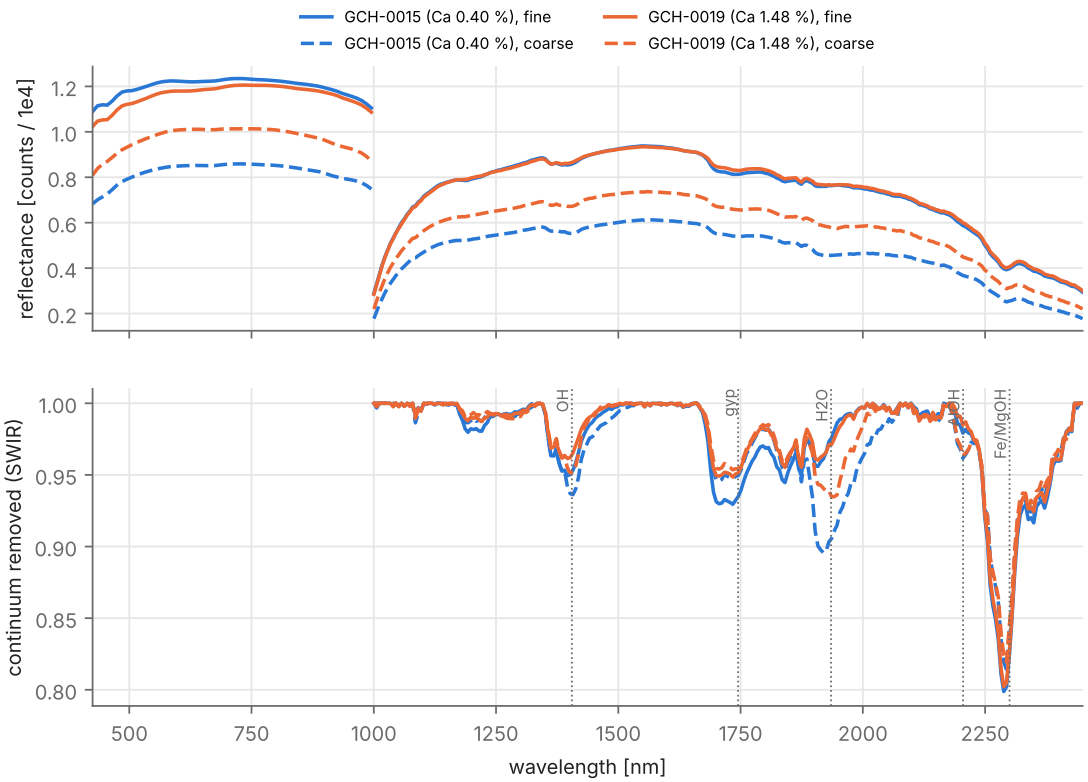


Figure 1: Mean spectra of the samples with the lowest and the highest Ca content. Top: reflectance of the two cameras; coarse crops (dashed) are darker. Bottom: SWIR spectrum divided by its convex hull (continuum removal, Clark and Roush, 1984). The difference between the fine and the coarse fraction of the *same* sample is larger than the difference between the two samples; in this pair the high-Ca sample even has the weaker water band, against the trend over all samples (Table 1).

Not every pixel is ore. Figure 3 maps the band depths pixel by pixel for two samples and clusters the continuum-removed SWIR spectra of all pixels with k-means. One cluster has absorptions at 1210, 1730 and 2310 nm: the C–H overtones of plastic, i.e. the sample tray, visible at the edges of the crops. Over all 106 crops, the tray makes up a median of 4 % and at most 17 % of the pixels. The other clusters separate the size fractions and follow vertical stripes of the push-broom sensor rather than individual particles: at the resolution of these cubes, the fine fraction cannot be resolved into single grains.

4 Method

Validation. All results are *leave-one-sample-out* (LOSO): each sample is predicted by a model trained on the other 27, and every hyper-parameter is chosen by an inner cross-validation grouped by sample on the training samples only. The predictions for the crops or patches of a test sample are averaged into one prediction per sample. The reference is the mean predictor, which predicts the mean composition of the training samples. We report the skill

$$\text{skill} = 1 - \frac{\text{MAE}}{\text{MAE}_{\text{mean predictor}}},$$

the fraction of the error of the mean predictor that the model removes (0: no better than the mean, 1: perfect), and the predictive Q^2, the analogous ratio of squared errors. The mean absolute percentage error (MAPE) is given for comparison with the literature, but it depends on the level of the element and says nothing without the reference of the mean predictor. Whether a

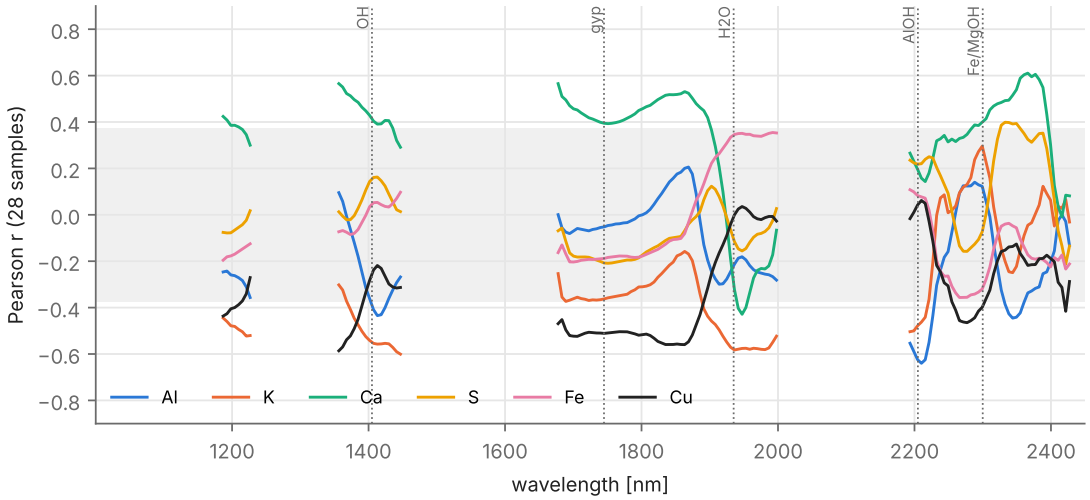


Figure 2: Correlation between continuum-removed SWIR reflectance and composition over the 28 samples. Negative values mean a deeper absorption in samples richer in the element. Bands where no sample absorbs (continuum ≈ 1) are omitted. Grey: $|r| < 0.37$, not significant at $p = 0.05$ for a single band.

Table 1: Band depths (Clark and Roush, 1984) of the main absorption features: correlation of the sample means with the composition, the spread between samples, and the mean difference between the coarse and the fine fraction of the same sample (25 samples with both fractions).

	correlation with element (28 samples)						variation	
feature	Al	K	S	Ca	Fe	Cu	between samples (sd)	coarse − fine
OH 1405	+0.43	+0.55	−0.16	−0.39	−0.04	+0.22	0.0057	+0.0123
gyp 1745	+0.08	+0.38	+0.38	+0.21	+0.09	+0.27	0.0007	+0.0004
H2O 1935	+0.25	+0.56	+0.13	+0.49	−0.35	−0.13	0.0091	+0.0316
AlOH 2205	+0.58	+0.52	−0.20	−0.14	−0.15	−0.09	0.0062	+0.0211
Fe/MgOH 2300	−0.40	−0.67	+0.31	−0.11	+0.29	+0.23	0.0021	−0.0016
red/blue 700/480	−0.20	−0.02	−0.16	+0.39	−0.27	−0.32	0.0129	+0.0222

skill is better than chance is tested by permuting the labels between whole samples and repeating the complete LOSO procedure 100 times (Ojala and Garriga, 2010).

Models. Partial least squares (PLS, Wold et al., 2001), the standard of chemometrics; principal component analysis followed by ridge regression (PCA + ridge); a re-implementation of the HIDSAG validation pipeline (Egaña et al., 2020), k-means clustering of patch spectra with one PLS per cluster; a small 1D convolutional network (three convolution layers, about 4000 weights, early stopping on held-out samples); and, as the most interpretable competitor, a single best band per element, re-selected in every fold. All models predict the 18 standardised elements jointly. Pre-processing (Rinnan et al., 2009): none (raw), standard normal variate (SNV), or SNV followed by a Savitzky–Golay first derivative, computed separately for each camera.

5 Results

Table 2 and Figure 4 summarise the benchmark. The best model, PCA + ridge on the raw crop means, removes 29 % of the error of the mean predictor on average. The elements fall into three groups:

- **Ca and Fe** (skill 0.50 and 0.49, Q^2 0.71 and 0.69): predictable; Ca through the water band, Fe through information spread over the whole spectrum rather than a single feature;

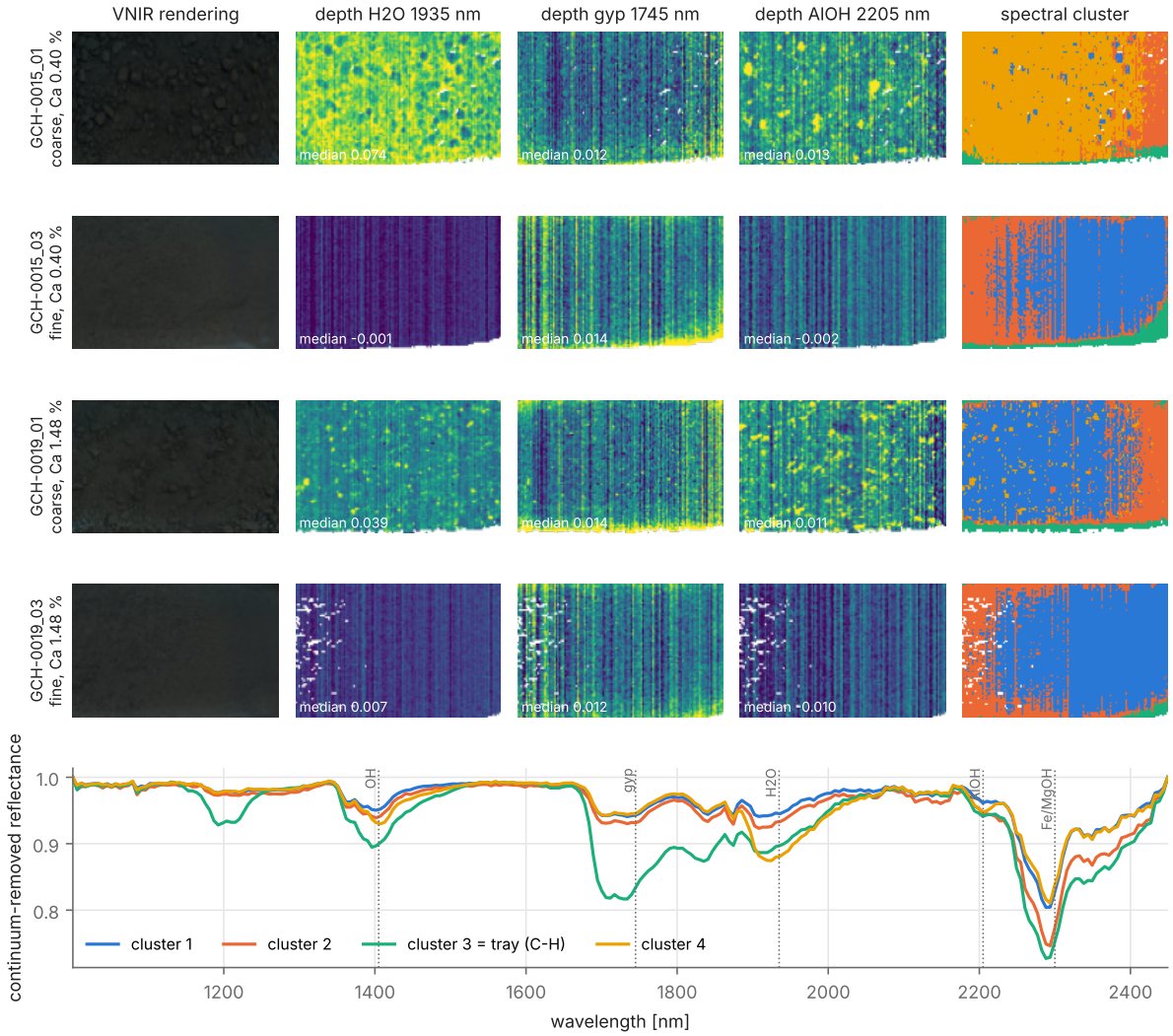


Figure 3: Pixel maps of a low-Ca (GCH-0015) and a high-Ca (GCH-0019) sample, coarse (01) and fine (03) crops. Columns: VNIR rendering, band depths at 1935, 1745 and 2205 nm (same colour scale per column), and k-means clusters of the continuum-removed SWIR spectra, with the cluster centres below. The tray cluster appears at the lower edge of the crops. The maps differ more between the two crops of one sample than between the two samples.

Table 2: Leave-one-sample-out benchmark: mean skill and Q^2 over the nine major elements, and skill per element. Patch level: 40 patches of 25 pixels per crop (4240 rows, still 28 samples).

model	level	pre-proc.	skill	Q^2	Al	K	S	Ca	Fe	Cu
mean	crop	raw	+0.00	−0.00	−0.01	+0.02	−0.00	−0.00	−0.00	−0.00
best band	crop	SNV + 1st deriv.	+0.07	+0.06	+0.16	+0.07	−0.24	+0.41	+0.20	+0.37
PLS	crop	raw	+0.20	+0.37	+0.18	+0.13	+0.20	+0.40	+0.47	+0.20
PLS	crop	SNV + 1st deriv.	+0.24	+0.40	+0.25	+0.03	+0.27	+0.48	+0.39	+0.30
PCA + ridge	crop	raw	**+0.29**	+0.48	+0.29	+0.33	+0.31	+0.50	+0.49	+0.28
PCA + ridge	crop	SNV + 1st deriv.	+0.27	+0.46	+0.22	+0.26	+0.31	+0.50	+0.38	+0.30
1D-CNN	crop	SNV	+0.02	+0.04	+0.07	+0.03	−0.01	−0.02	+0.12	+0.02
PLS	patch	SNV	+0.26	+0.44	+0.23	+0.26	+0.31	+0.52	+0.32	+0.26
PCA + ridge	patch	SNV	+0.23	+0.39	+0.20	+0.10	+0.30	+0.49	+0.29	+0.25
cluster + PLS (HIDSAG-like)	patch	SNV	+0.22	+0.39	+0.14	+0.23	+0.38	+0.41	+0.27	+0.31
1D-CNN	patch	SNV	+0.17	+0.32	+0.14	+0.22	+0.10	+0.31	+0.21	+0.17

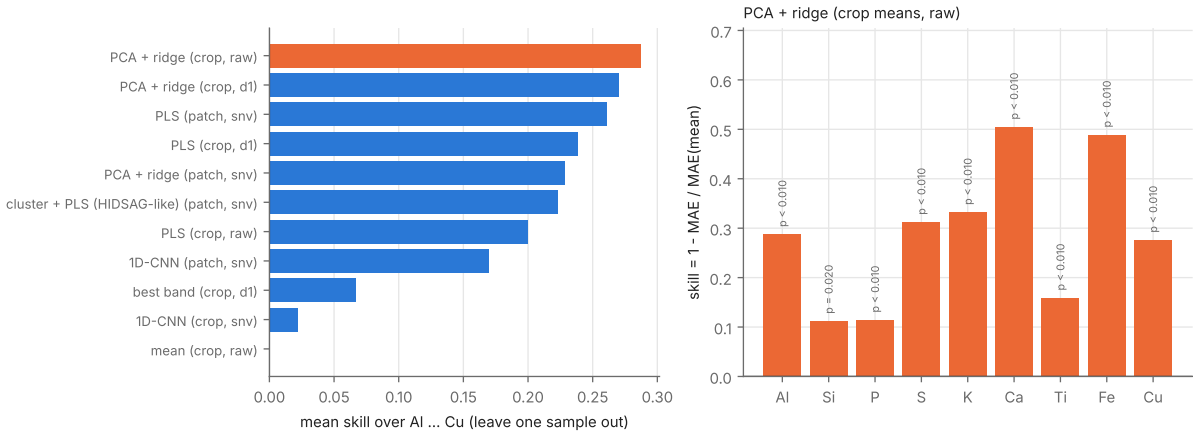


Figure 4: Left: mean skill of all configurations (Al . . . Cu). Right: skill of the best model per element, with the p-value of the permutation test (100 permutations of the labels between samples).

- **K, S, Al, Cu** (0.28–0.33): partly predictable, through muscovite (Al, K) and through the association of sulfides with alteration (S, Cu; S, Fe and Cu correlate with each other at $r \approx 0.55$);
- **Si, P, Ti** (0.11–0.16): hardly predictable; quartz, apatite and rutile have no absorptions in this range.

The permutation test (Table 3) confirms that the skill is not an artefact of the small sample: for PCA + ridge, no permutation of the labels reaches the observed skill for eight of the nine elements ($p < 0.01$), and for Si $p = 0.02$. Even the small skills of Si, P and Ti are real, if of little practical use. The predictions (Figure 5) are shrunk towards the mean, as expected from a regularised model trained on 27 points; the ranking of the samples is better than the absolute values.

More data per sample does not help. Patch-level models see 40 times more rows, but they do not do better than crop-level models (Table 2): the extra rows describe the variability within a sample, which the label does not resolve. The HIDSAG-like cluster pipeline is not better than a single PLS. The 1D-CNN is the weakest learned model: barely better than the mean predictor on crop means (skill +0.02) and +0.17 on patches, where it has more rows to fit, still below PLS. With 28 independent labels, the bias of a linear model is a better trade than the variance of a network.

Table 3: Permutation test: LOSO skill and its p-value under 100 random re-assignments of the compositions to the samples (crop level).

model		Al	Si	P	S	K	Ca	Ti	Fe	Cu
PCA + ridge (raw)	skill	+0.29	+0.11	+0.11	+0.31	+0.33	+0.50	+0.16	+0.49	+0.28
	p	<0.01	0.020	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
PLS (SNV + 1st deriv.)	skill	+0.25	+0.11	+0.17	+0.27	+0.03	+0.48	+0.15	+0.39	+0.30
	p	<0.01	0.020	<0.01	<0.01	0.109	<0.01	<0.01	<0.01	<0.01

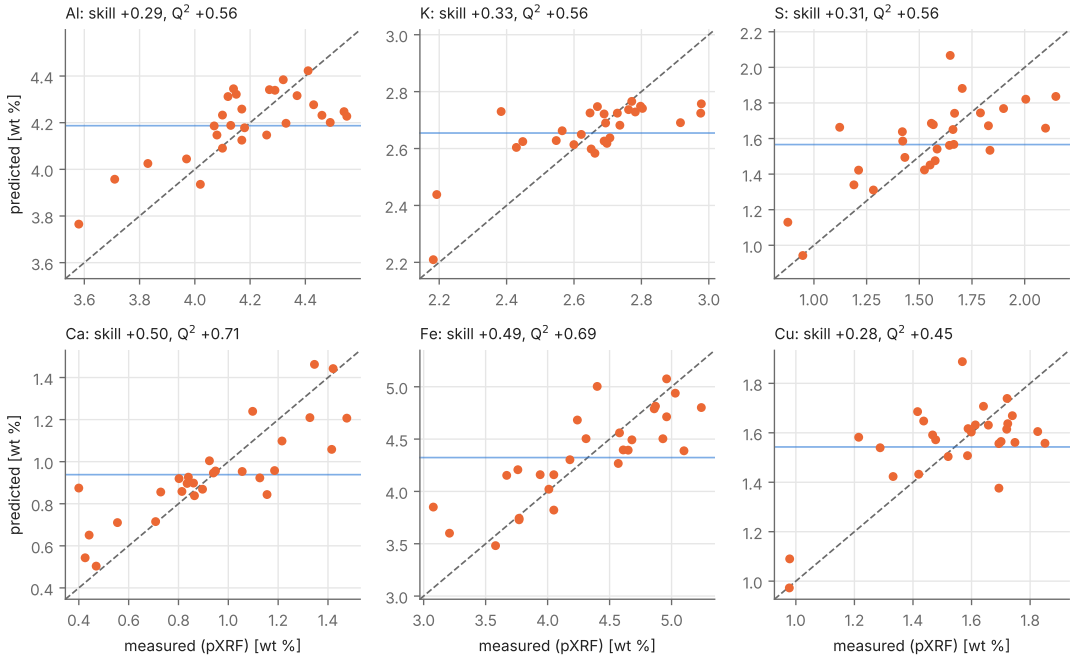


Figure 5: Predicted against measured composition (LOSO, PCA + ridge, crop means). Dashed: identity; blue: mean of the samples.

Corrections that did not help. Table 4 tests whether the artefacts of Section 3 hurt the model. Removing the tray pixels from the crop means does not change the skill, nor does centring the spectra per size fraction; the linear models already learn to ignore the grain-size direction when both fractions are in the training set, and the centring removes some useful information. The fine fraction alone is more informative than the coarse one, and using both is best.

6 Comparison with the reference benchmark

The HIDSAG paper reports for GEOCHEM a mean absolute percentage error of 0.021 % for Al and below 0.3 % for all nine major elements with the full spectral range, obtained with "leave-one-out rounds" of a clustering and regression pipeline (Ehrenfeld et al., 2023a, Table 5). Table 5 puts these numbers next to ours. The mean predictor alone has a MAPE of 4.5 % for Al; our best LOSO model reaches 3.1 %.

To find out which validation can produce the published numbers, Table 6 and Figure 6 evaluates the same models with three splits of the patch-level data: random patches (80/20), leave one crop out, and leave one sample out. Random patch splits put patches of every test sample in the training set; they inflate the skill and bring the MAPE close to the SWIR-only column of the paper. No split brings it near the full-range column, which is two orders of magnitude below the SWIR column of the same table. We conclude that the published GEOCHEM numbers were most likely obtained with a split that shares samples between training and test, and that the full-range

Table 4: Robustness checks (LOSO, mean skill over Al … Cu and skill of three elements).

variant	model	samples	skill	Ca	Fe	Cu
baseline	PCA + ridge (raw)	28	+0.29	+0.51	+0.49	+0.28
baseline	PLS (d1)	28	+0.24	+0.48	+0.39	+0.30
grain-size centring	PCA + ridge (raw)	28	+0.21	+0.40	+0.46	+0.21
grain-size centring	PLS (d1)	28	+0.18	+0.42	+0.39	+0.32
both fractions (25 samples)	PCA + ridge (raw)	25	+0.25	+0.44	+0.52	+0.37
both fractions (25 samples)	PLS (d1)	25	+0.22	+0.56	+0.38	+0.35
fine only (25 samples)	PCA + ridge (raw)	25	+0.18	+0.09	+0.48	+0.45
fine only (25 samples)	PLS (d1)	25	+0.19	+0.50	+0.18	+0.31
coarse only (25 samples)	PCA + ridge (raw)	25	+0.05	+0.18	+0.19	+0.25
coarse only (25 samples)	PLS (d1)	25	+0.12	+0.51	+0.35	+0.32
1000 px per crop	PCA + ridge (raw)	28	+0.28	+0.49	+0.51	+0.27
1000 px per crop	PLS (d1)	28	+0.22	+0.47	+0.40	+0.30
1000 px, tray masked	PCA + ridge (raw)	28	+0.26	+0.46	+0.50	+0.28
1000 px, tray masked	PLS (d1)	28	+0.23	+0.49	+0.39	+0.25

Table 5: MAPE [%] of the major elements: mean predictor and best model under LOSO, the HIDSAG-like pipeline under a random patch split, and the values reported by Ehrenfeld et al. (2023a, Table 5).

	this work			HIDSAG paper	
element	mean pred.	LOSO	random patches	SWIR	full
Al	4.5	3.1	2.7	2.5	0.021
Si	3.2	2.9	2.2	1.7	0.136
P	7.2	6.3	4.9	3.9	0.181
S	17.5	11.1	7.5	6.8	0.168
K	5.5	3.7	3.2	2.4	0.080
Ca	35.3	16.6	14.5	7.5	0.022
Ti	11.7	9.8	7.7	6.9	0.213
Fe	12.4	6.3	6.3	4.7	0.152
Cu	13.2	8.7	6.6	5.8	0.270

column in addition contains an error of units or scale. They should not be used as a benchmark.

Under the same sample-wise protocol, the comparison is in favour of simpler models: our approximation of the reference pipeline (cluster + PLS) reaches a skill of 0.22 on patches, a single PLS 0.26 and PCA + ridge on crop means 0.29 (Table 2). The approximation differs from the original in its details (the original uses a probabilistic clustering, stochastic patch selection and ordinary least squares), so this ranking holds for our re-implementation, not necessarily for the authors' code.

7 Main changes

Table 7 lists the main modifications with respect to the original course notebook.

Table 6: The same models under three splits of the patch-level data. Skill: mean over Al . . . Cu.

model	split	skill	MAPE [%]: Al, Ca, Cu		
PLS	random patches (80/20)	+0.46	2.43	12.5	6.6
PLS	leave one crop out	+0.38	2.78	13.6	7.6
PLS	leave one sample out	+0.26	3.38	15.4	9.2
cluster + PLS (HIDSAG-like)	random patches (80/20)	+0.43	2.70	14.5	6.6
cluster + PLS (HIDSAG-like)	leave one crop out	+0.33	3.20	16.9	7.5
cluster + PLS (HIDSAG-like)	leave one sample out	+0.22	3.79	19.1	8.6

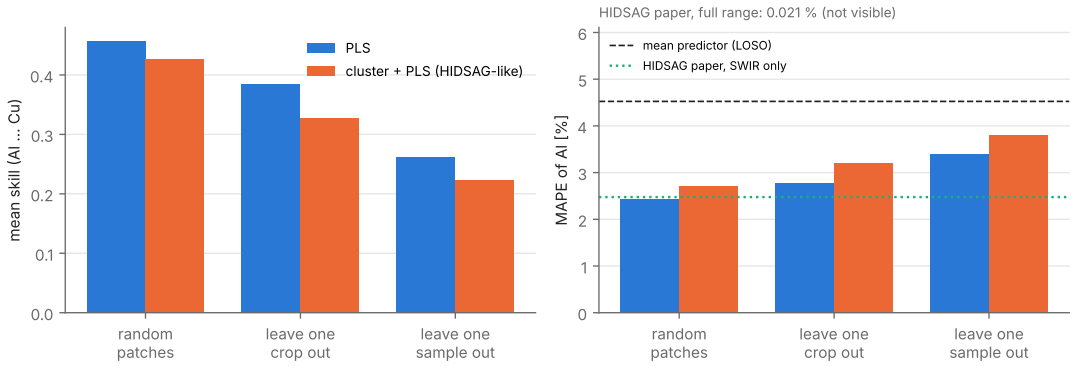


Figure 6: Effect of the split on the measured accuracy. Left: mean skill. Right: MAPE of Al; dashed: mean predictor, dotted: value reported by the HIDSAG paper for the SWIR camera alone. Only the right-most pair measures the prediction of an unseen sample.

Table 7: Main changes with respect to the original notebook.

	original notebook	this version
Data used	one mean spectrum per sample (28 spectra)	all 106 crops, both size fractions, 1000 pixels per crop; patches of 25 pixels
Spectral features	–	VNIR binned to 10 nm, noisy band edges removed; SNV, derivatives, continuum removal, band depths
Validation	random split	leave one sample out, inner cross-validation grouped by sample, permutation test
Reference	none	mean predictor; skill and Q^2 relative to it
Models	1D-CNN	PLS, PCA + ridge, cluster + PLS (reference pipeline), best band, 1D-CNN
Interpretation	–	absorption features, correlation spectra, pixel maps
Artefacts	not examined	grain size, tray pixels, sensor striping quantified and tested
Comparison with literature	published accuracy taken at face value	same models under leaky and honest splits
Report	Jupyter notebook	this report, scripts for every table and figure, tests

8 Further work and conclusions

Conclusions.

1. Hyperspectral images of a sampled batch carry real but limited information about its

chemistry. On samples not seen in training, the best model removes 29 % of the error of the mean predictor on average, about half for Ca and Fe, about a third for K, S, Al and Cu, and little for Si, P and Ti. All skills are significant in a permutation test.

2. The camera measures minerals. Elements are predictable when they reside in minerals with absorptions in the VNIR–SWIR range (Al and K in muscovite) or co-vary with them (Cu with alteration).

3. With 28 labelled samples, regularised linear models are the right tools; a convolutional network and a clustering pipeline add variance, not skill. More pixels per sample do not compensate for few samples.

4. Grain size changes the absorption depths more than composition does. Sample preparation should fix the size fraction, or the size distribution has to be measured alongside the spectra.

5. The accuracy published for HIDSAG GEOCHEM is consistent with a validation that shares samples between training and test, and in part with an error of scale. It cannot serve as a target.

Further work. For the SAG mill, what matters is not the composition itself but hardness, grindability and moisture. The HIDSAG subset GEOMET contains 146 drill-core samples with the Bond work index, copper and molybdenum recoveries and other metallurgical tests, imaged with the same cameras. There, the reference pipeline was validated on 21 samples held out from training (Ehrenfeld et al., 2023a), i.e. by sample: its GEOMET results are a legitimate baseline, and the natural starting point of the next phase, unlike its GEOCHEM results. Three lessons carry over from this work: validate by sample (or by drill hole), report the skill with respect to the mean predictor, and model the particle size. Whether the predicted composition should be an intermediate input to the mill variables, or the spectra should be regressed on them directly, is an open question: the errors of this first step would propagate into the second, which favours the direct route unless the composition adds information.

Data and code

HIDSAG is published under CC BY 4.0 (Ehrenfeld et al., 2023b). All code, the derived crop spectra and the scripts for every table and figure are available at Chavarría Morales (2026).

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